Metagenomic Profiling

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Introduction

Shotgun metagenomics sequences all DNA in a sample rather than a marker gene, enabling species-level taxonomic and functional profiling. Two dominant approaches: MetaPhlAn (marker-gene alignment to species-specific genes) and Kraken (k-mer matching against a reference database).

Prerequisites

Shotgun sequencing basics; read QC; familiarity with amplicon methods.

Theory

MetaPhlAn maps reads to ~1M clade-specific markers; species abundances are inferred as relative marker coverage. Output is a per-species relative abundance table.

Kraken2 classifies reads via k-mer matches to a database (e.g., RefSeq Bacteria). Fast and comprehensive but prone to false positives for low-abundance species; pair with **Bracken** for abundance re-estimation.

Functional profiling: HUMAnN3 for pathway-level abundances (MetaCyc); eggNOG-mapper for gene-function annotation.

Assumptions

Reads are adapter-trimmed and host-decontaminated; reference database covers the expected taxa; relative abundances are interpretable only within a sample.

R Implementation

MetaPhlAn and Kraken run on the command line; R ingests the results for downstream diversity and DA analysis:

```r
library(phyloseq)

set.seed(2026)
# Simulated MetaPhlAn-style output
species <- paste0("s__Species_", 1:50)
samples <- paste0("S", 1:20)
abund <- matrix(rgamma(length(species) * length(samples), 0.5, 2),
                length(species), length(samples))
abund <- sweep(abund, 2, colSums(abund), "/")   # relative abundances
rownames(abund) <- species; colnames(abund) <- samples

# Coerce to a phyloseq object for diversity analyses
tax_tab <- matrix(species, length(species), 1,
                  dimnames = list(species, "Species"))
ps <- phyloseq(otu_table(round(abund * 1e6), taxa_are_rows = TRUE),
               tax_table(tax_tab))
```

```r
ps

# Top species by mean abundance
head(sort(rowMeans(abund), decreasing = TRUE), 5)
```

Example command-line steps (not run here):

```
metaphlan sample.fastq --input_type fastq --bowtie2db metaphlan_db \
          -o sample_profile.txt
kraken2 --db kraken_db --paired R1.fq R2.fq --report sample.kreport
```

Output & Results

A species-by-sample abundance table; in phyloseq, all standard diversity and DA methods apply.

Interpretation

"MetaPhlAn profiling revealed 420 species; top differentially abundant species between gut inflammation cases and controls included Faecalibacterium prausnitzii (depleted) and Escherichia coli (enriched)."

Practical Tips

- Host-read removal (Bowtie2 vs the host genome) is essential; residual host DNA biases classification.
- Kraken2 + Bracken gives per-species quantification with reduced false positives vs Kraken alone.
- For functional analysis, HUMAnN3 provides pathway abundances suitable for differential pathway analysis with MaAsLin2 or LEfSe.
- ONT long reads enable better species resolution in complex communities.
- Do not mix relative and absolute abundances across analyses; compositional data require log-ratio transforms (CLR) or specific methods (ANCOM-BC).

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat